

THE S-COMPLEX DIFFERENTIAL OF TORUS KNOTS

BERND JOHANNES WUEBBEN

ABSTRACT. For a torus knot $K = T(p, q)$ the traceless $\mathrm{SU}(2)$ character variety is finite and nondegenerate, so Daemi and Scaduto's S-complex for singular instanton homology can be built on it. The resulting complex has rank $1 + |\sigma(K)|$, where σ is the signature, and by theorems of Li and Ye and of Kronheimer and Mrowka its homology $I^\natural(K)$ has rank $\|\Delta_K\|_1$, the sum of the absolute values of the coefficients of the Alexander polynomial; so the differential has rank $R(p, q) = \frac{1}{2}(1 + |\sigma| - \|\Delta\|_1)$. We express $R(p, q)$ through floor sums and show that, for coprime $2 \leq p < q$, $R(p, q) \geq \lfloor (p - 2b)^2/4 \rfloor$, where $b \in \{1, \dots, p - 1\}$ is the inverse of q modulo p , and that the differential vanishes if and only if $p = 2$, or p is odd and $q \equiv 2$ or $q \equiv p + 2(-1)^{(p-1)/2} \pmod{2p}$; the 'if' direction is due to Daemi and Scaduto. By a result of Daemi and Scaduto, the component of the differential to the reducible generator is nonzero over $\mathbb{Q}$ exactly when their invariant h is positive. We show that the components from degree 3 to degree 2 and from degree 1 to degree 0 have ranks $\lfloor R/2 \rfloor$ and $\lceil R/2 \rceil$, independently of the choices in the construction. In particular the $\mathbb{Z}/4$ -graded group $I^\natural(T(p, q); \mathbb{Q})$ depends only on σ and $\|\Delta\|_1$, and we give it explicitly. In each small-perturbation pillowcase computation of Hedden, Herald and Kirk, a bigon ends at the reducible generator exactly when $h > 0$. We formulate a chain-level conjecture identifying the two complexes over $\mathbb{F}_2$; it predicts this pattern whenever the map to the reducible is nonzero modulo 2.

1. INTRODUCTION

1.1. Background. Kronheimer and Mrowka's reduced singular instanton homology $I^\natural(K)$ of a knot $K \subset S^3$ is built from traceless $\mathrm{SU}(2)$ representations of $\pi_1(S^3 \setminus K)$, those sending every meridian to a conjugate of $\mathrm{diag}(i, -i)$ [KM11a, KM11b]. Daemi and Scaduto refined the underlying chain complex to an *S-complex* [DS24b, Def. 3.21]

$$\tilde{C}_*(K) = C_*(K) \oplus C_{*-1}(K) \oplus \mathbb{Z}, \quad \tilde{d} = \begin{pmatrix} d & 0 & 0 \\ v & -d & \delta_2 \\ \delta_1 & 0 & 0 \end{pmatrix}.$$

Here $C_*(K)$ is generated by the irreducible critical points of the singular Chern–Simons functional, and the summand $\mathbb{Z}$, in degree 0, by the reducible (abelian) one. The maps δ_1 and δ_2 count instantons to and from the reducible, and the coefficients of $v: C_* \rightarrow C_{*-2}$ are the degrees of holonomy maps to the circle on one-dimensional moduli spaces of instantons between irreducibles [DS24b, §3.3, (3.15)]. Gradings are homological, differentials have degree -1 , and degrees are read modulo 4. The S-complex structure is the map $\chi: \tilde{C}_* \rightarrow \tilde{C}_{*+1}$, $\chi(\alpha, \beta, n) = (0, \alpha, 0)$, which satisfies $\chi^2 = 0$ and $\chi\tilde{d} + \tilde{d}\chi = 0$ [DS24b, §4.1]. A *morphism* of S-complexes is a chain map of degree 0 that commutes with χ and induces the identity on the reducible summand [DS24b, Def. 3.28, §4.1]; an *S-isomorphism* is an invertible morphism; and an *S-chain homotopy equivalence* is a pair of morphisms whose composites are chain homotopic to the identity through homotopies that anticommute with χ [DS24b, Def. 3.30, §4.1]. The S-chain homotopy type of $\tilde{C}(K)$ depends

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only on the oriented knot K [DS24b, Thm. 3.34, Rem. 3.35], its homology is $I^\natural(K)$ up to a shift of grading [DS24b, Thm. 8.9], and its local equivalence class carries Frøyshov-type concordance invariants, among them the integer $h(K)$ [DS24b, §5]. Daemi and Scaduto also define an equivariant homology $\hat{I}(K)$, a module over $\mathbb{Z}[x]$ [DS24b, §4.3, Thm. 1.4].

For a torus knot $K = T(p, q)$ the traceless character variety is finite and nondegenerate, with $|\sigma(K)|/2$ irreducible points and one abelian point [HHK14, §5.1], [DS24b, §9.5]. The S-complex can therefore be formed with a holonomy perturbation that is trivial near this critical set, with one generator for each irreducible character and one for the abelian character [DS24a, §4.5, proof of Thm. 4]; we call such an S-complex *built on the character variety*. The irreducible generators lie in degrees 1 and 3 [DS24a, §1, Cor. 3 and its proof], so $d = 0$, $C_*(K)$ has rank $|\sigma|/2$, and $\tilde{C}(K)$ has rank $1 + |\sigma|$. Li and Ye showed that Kronheimer and Mrowka's instanton knot homology $KHI(K)$ of a torus knot has dimension $\|\Delta_K\|_1$, the sum of the absolute values of the coefficients of the Alexander polynomial [LY22, Cor. 1.8]; by Kronheimer and Mrowka's isomorphism $I^\natural(K; \mathbb{Q}) \cong KHI(K; \mathbb{Q})$ [KM11b, Prop. 1.4], the homology of $\tilde{C}(K; \mathbb{Q})$ has the same rank. Hence the rank of the S-complex differential of $T(p, q)$ is

$$R(p, q) := \text{rank}_{\mathbb{Q}} \tilde{d} = \frac{1}{2}(1 + |\sigma(T(p, q))| - \|\Delta_{T(p, q)}\|_1)$$

(Proposition 2.1); this consequence is also recorded, with its values for $T(3, n)$, in [Wue26a, Cor. 1.4]. This raises two questions. For which torus knots does the differential vanish, so that the unperturbed singular Chern–Simons functional is perfect? And which components of $\tilde{d}$, the map δ_1 to the reducible or the map v between irreducibles, carry the differential when it does not vanish?

Daemi and Scaduto answered the first question in one direction: they showed that $1 + |\sigma| = \|\Delta\|_1$, and hence $\tilde{d} = 0$ and $h = 0$, for two explicit families of torus knots [DS24b, (9.36), Prop. 9.37]. They also computed $T(3, 4)$ and $T(3, 5)$, where $\delta_1 \neq 0$ [DS24b, §§9.3–9.4]. Hedden, Herald and Kirk computed the perturbed traceless character varieties of several torus knots. Comparing Khovanov homology with the Alexander polynomial, they determined the $\mathbb{Z}/4$ -graded group $I^\natural(T(3, n))$ for $n \leq 38$, whose rank is $1 + |\sigma|$ or $|\sigma| - 1$, so that the differential has rank at most one [HHK14, §12.4]. They also noted that Kronheimer and Mrowka's conjectured isomorphism between KHI and knot Floer homology [KM10b, Conj. 7.25] would force the differential to have large rank in general [HHK14, §12.6]. With Li and Ye's theorem the rank of the differential is an explicit function of σ and Δ , and Theorem 1.1 below confirms this expectation.

1.2. Results. Throughout, $p, q \geq 2$ are coprime, $T(p, q) = T(q, p)$ is the positive torus knot, and signatures are normalized so that positive torus knots have $\sigma < 0$; where needed we also allow $q = 1$, for which $T(p, 1)$ is the unknot and $R(p, 1) = 0$. For a statement $\mathcal{S}$ we write $[\mathcal{S}] = 1$ if $\mathcal{S}$ holds and $[\mathcal{S}] = 0$ otherwise. For $p \geq 2$ let $b = b_p(q) \in \{1, \dots, p-1\}$ be the inverse of q modulo p , and put

$$u_p(q) = p - 2b_p(q), \quad A_j = \left\lfloor \frac{jq}{p} \right\rfloor, \quad F(n) = \sum_{j=0}^{n-1} A_j, \quad m(p, q) = \sum_{\substack{1 \leq k \leq p-2 \\ k \equiv p \pmod{2}}} \left\lfloor \frac{kq}{2p} \right\rfloor;$$

when the pair matters we write $F_{p,q}$. Since $T(p, q) = T(q, p)$, we have $R(p, q) = R(q, p)$, although the formula below is not symmetric in p and q .

Theorem 1.1 (Floor-sum formula and lower bound). *For coprime $p \geq 2$ and $q \geq 1$,*

$$R(p, q) = F(b) + F(p - b) - 2m(p, q) \quad \text{and} \quad R(p, q + 2p) = R(p, q) + \left\lfloor \frac{u_p(q)^2}{2} \right\rfloor.$$

If moreover $p < q$, then

$$R(p, q) \geq \left\lfloor \frac{u_p(q)^2}{4} \right\rfloor.$$

We call the second identity the *step law*. The quantity $|u_p(q)|$ measures how far b is from $p/2$. In terms of the semigroup $\langle p, q \rangle$, R compares the counting function of the semigroup at two points with its value at their midpoint, and the bound says that R is small only when the two points are nearly symmetric (Remark 3.1). The bound is sharp: for even $p \geq 4$, $b_p(p+1) = 1$ and $R(p, p+1) = ((p-2)/2)^2 = u_p(p+1)^2/4$. In particular R is unbounded, as Hedden, Herald and Kirk anticipated [HHK14, §12.6]. The smallest torus knot with $R \geq 2$, by genus and by crossing number, is $T(4, 7)$, where $R = 2$; for $T(5, 11)$, $R = 4$.

Theorem 1.2 (Vanishing of the differential). *Let $2 \leq p < q$ be coprime. The S -complex differential of $T(p, q)$ vanishes, equivalently $\text{rank}_{\mathbb{Q}} I^{\natural}(T(p, q)) = 1 + |\sigma(T(p, q))|$, equivalently $\Delta_{T(p, q)}$ has exactly $1 + |\sigma(T(p, q))|$ nonzero coefficients, if and only if one of the following holds:*

- (1) $p = 2$;
- (2) p is odd and $q \equiv 2 \pmod{2p}$;
- (3) $p \equiv 1 \pmod{4}$ and $q \equiv p + 2 \pmod{2p}$, or $p \equiv 3 \pmod{4}$ and $q \equiv p - 2 \pmod{2p}$.

The torus knots in (2) and (3) are exactly the two families of [DS24b, Prop. 9.37], so the new content of Theorem 1.2 is the converse: no other torus knot has a perfect unperturbed Chern–Simons functional. For $p = 3$ it says that the differential of $T(3, n)$ vanishes exactly when $n \equiv 1, 2 \pmod{6}$, in agreement with the computations of Hedden, Herald and Kirk for $n \leq 38$ [HHK14, §12.4]. For torus knots $h(K) = \Upsilon_K(1) - \sigma(K)/2$ [DS24a, §1, Prop. 1 and the paragraph following it], where Υ_K is the invariant of Ozsváth, Stipsicz and Szabó [OSS17], normalized so that $\Upsilon_{T(2,3)}(1) = -1$. Jabuka and Van Cott showed that the knots in (1)–(3) are exactly the torus knots with $\Upsilon_K(1) = \sigma(K)/2$ that become the unknot after a single pinch move of Batson [Bat14] (a nonorientable band move) [JVC21, Prop. 1.6, Thms. 1.7–1.8]. Theorem 1.2 can therefore be read as: the differential of a torus knot vanishes if and only if $h = 0$ and one pinch move unknots it.

For the second question, let $N_1 = \lceil |\sigma|/4 \rceil$ and $N_3 = \lfloor |\sigma|/4 \rfloor$. The complex $C_*(K)$ is $C_1 \oplus C_3$, with $\text{rank } C_1 = N_1$ and $\text{rank } C_3 = N_3$ [DS24a, §1, Cor. 3]. Since v has degree -2 , it exchanges C_1 and C_3 ; write $v = v_{13} + v_{31}$ with $v_{13}: C_1 \rightarrow C_3$ and $v_{31}: C_3 \rightarrow C_1$. In $\tilde{C}$ the group C_1 appears in degrees 1 and 2, and C_3 in degrees 3 and 0; thus v_{31} is the component of $\tilde{d}$ from degree 3 to degree 2, while δ_1 and v_{13} together form the component from degree 1 to degree 0. The map v itself is not an invariant of K : an S -isomorphism can add to it a map of rank one (Remark 7.3).

Theorem 1.3 (Components of the differential). *Let $K = T(p, q)$, and use rational coefficients. For $w \in C_3$ write $w\delta_1$ for the map $c \mapsto \delta_1(c)w$.*

- (1) *Any two S -complexes of K built on its character variety are S -isomorphic. Every S -isomorphism from $\tilde{C}$ to $\tilde{C}'$ is given by a pair (P, w) , with $P: C \rightarrow C'$ an isomorphism preserving degrees and $w \in C'_3$, and then $v' = PvP^{-1} + w\delta_1P^{-1}$ and $\delta'_1 = \delta_1P^{-1}$.*
- (2) [DS24b, Prop. 4.15] $\delta_1 \neq 0$ if and only if $h(K) > 0$.
- (3) $\text{rank } v_{31}$ and $\text{rank}(v_{13}|_{\ker \delta_1})$ do not depend on the choice of S -complex built on the character variety, and

$$\text{rank } v_{31} = \left\lfloor \frac{R(p, q)}{2} \right\rfloor, \quad \text{rank}(v_{13}|_{\ker \delta_1}) = \left\lceil \frac{R(p, q)}{2} \right\rceil - [h(K) > 0];$$

equivalently, the components of $\tilde{d}$ from degree 3 to degree 2 and from degree 1 to degree 0 have ranks $\lfloor R(p, q)/2 \rfloor$ and $\lceil R(p, q)/2 \rceil$.

- (4) If $h(K) = 0$, then $\delta_1 = 0$, and v is an endomorphism of $C_*(K; \mathbb{Q})$ of rank $R(p, q)$, well defined up to conjugation: it is the action of x on the $\mathbb{Q}[x]$ -torsion of Daemi and Scaduto's equivariant homology $\widehat{I}(K; \mathbb{Q})$.

Corollary 1.4. *Let $K = T(p, q)$ and $R = R(p, q)$. In the grading of the homology of $\widetilde{C}(K; \mathbb{Q})$, in which the reducible generator has degree 0, $I^l(K; \mathbb{Q})$ has rank vector*

$$(N_3 + 1 - \lceil R/2 \rceil, N_1 - \lceil R/2 \rceil, N_1 - \lfloor R/2 \rfloor, N_3 - \lfloor R/2 \rfloor)$$

in degrees 0, 1, 2, 3. The absolute grading of I^l used in [DS24b, Thm. 8.9], that of [KM11b, §4.5], is obtained from this one by adding $\sigma(K)$ modulo 4.

Earlier determinations of the $\mathbb{Z}/4$ -graded group include $T(4, 5)$ [KM14, §11], $T(3, n)$ with $n \leq 38$ [HHK14, §12.4], and, since $\widetilde{d} = 0$ there, the two families of [DS24b, Prop. 9.37].

Item (1) of Theorem 1.3 follows from Daemi and Scaduto's definitions, item (2) is their result, and the identification in (4) is formal. The new content of Theorem 1.3 is (3), together with its consequences: the rank of v in (4) and Corollary 1.4. It rests on Lemma 7.1. Kronheimer and Mrowka's operator $\mu(\bar{\Sigma})$ on $KHI(K)$, whose generalized eigenspaces are the summands in Alexander grading j , has degree 2 modulo 4, and these summands have Euler characteristics equal to the coefficients of Δ_K up to sign [KM10a, Lim10]. With Li and Ye's computation of their dimensions this gives $\{\beta_0 - \beta_2, \beta_1 - \beta_3\} = \{0, \pm 1\}$, where β_l is the rank of $I^l(K; \mathbb{Q})$ in degree l . Since $\widetilde{C}$ has ranks $(N_3 + 1, N_1, N_1, N_3)$, the ranks of the components of degrees $3 \rightarrow 2$ and $1 \rightarrow 0$ then differ by at most one, and they sum to R . Kronheimer and Mrowka gave this argument for $T(4, 5)$ [KM14, §11].

Item (4) produces torus knots with $\delta_1 = 0$ and $v \neq 0$ over $\mathbb{Q}$. Nonzero maps v for torus knots were known before in two settings. (i) Daemi and Scaduto also use local coefficients, over rings containing a variable that they call T ; when $T^4 - 1$ is invertible, the S-complex of a knot is S-chain homotopy equivalent to that of the torus knot $T(2, |\sigma| + 1)$ or of its mirror, according to the sign of σ [DS24a, §1, Thm. 10], and over the field $\text{Frac}((\mathbb{Z}/r)[T^{\pm 1}])$, r prime, the S-complex of $T(p, q)$ has $\delta_1 v^{|\sigma|/2-1} \neq 0$ [DS24a, §4.5, proof of Thm. 4], so $v \neq 0$ once $|\sigma| \geq 4$. (ii) Over $\mathbb{Z}$, $\delta_1 v \neq 0$ whenever $h \geq 2$ [DS24b, Prop. 4.15], for instance for $T(5, 6)$ [DS24a, §4.1, Cor. 4]. The smallest torus knot with $\delta_1 = 0$ and $v \neq 0$ over $\mathbb{Q}$, by genus and by crossing number, is $T(4, 5)$, with $\sigma = -8$, $\|\Delta\|_1 = 7$, $R = 1$ and $h = 0$: its S-complex has $\delta_1 = 0$, $v_{31} = 0$ and v_{13} of rank one (Example 7.2). Other examples are $T(4, 9)$ ($R = 2$), $T(6, 7)$ ($R = 4$) and $T(5, 11)$ ($R = 4$). For $T(3, n)$ the differential is nonzero exactly when $n \equiv 4, 5 \pmod{6}$ (Corollary 6.3); then $h = 1$, and up to S-isomorphism over $\mathbb{Q}$ the differential is δ_1 alone (Remark 7.4).

1.3. The pillowcase. Hedden, Herald and Kirk computed pillowcase Floer homology for torus knots from a tangle decomposition in which one tangle is trivial and carries the earring [HHK18, §§10–11]. For small holonomy perturbations the pillowcase complex, over $\mathbb{F}_2$, has $1 + |\sigma|$ generators: a generator r_+ , the perturbation of the abelian character, in grading $\sigma \bmod 4$, and two generators x_i^- and x_i^+ for each irreducible traceless character, with $\text{gr } x_i^+ = \text{gr } x_i^- + 1$ (labels as in [HHK18, §11]). Thus both complexes have one generator for the abelian character and two, in adjacent gradings, for each irreducible character. If every bigon runs from some x_i^- either to r_+ or to some x_j^+ with $j \neq i$, then $\chi(x_i^-) = x_i^+$ and $\chi(x_i^+) = \chi(r_+) = 0$ make the pillowcase complex an S-complex $\mathcal{P}(K)$ with reducible generator r_+ , in which x_i^- and x_i^+ play the roles of the two copies of an irreducible generator. Comparing the computations of Hedden, Herald and Kirk with Theorem 1.3 suggests the following.

Conjecture 1.5. *Let $K = T(p, q)$, and consider the pillowcase complex of [HHK18, §§10–11] over $\mathbb{F}_2$, for the decomposition in which the earring lies on the trivial tangle and the integers r, s*

with $pr + qs = 1$ satisfy $0 < r < q$, with zero bounding cochains and a holonomy perturbation small enough that the complex has exactly the $1 + |\sigma|$ generators r_+ , $x_i^\pm$ described above.

- (a) Every bigon runs from some x_i^- either to r_+ or to some x_j^+ with $j \neq i$.
- (b) The S-complex $\mathcal{P}(K)$, with σ subtracted from its grading, is S-chain homotopy equivalent to $\tilde{C}(K; \mathbb{F}_2)$.

The decomposition of [HHK18, §10] depends on the choice of (r, s) . The condition $0 < r < q$ determines it, since $pr \equiv 1 \pmod{q}$, and all the choices printed in [HHK18, §11] satisfy it. The restrictions are needed. For larger perturbations the complex of $T(3, 4)$ becomes perfect, and with the earring on the other tangle the homology of $T(4, 5)$ has the wrong rank (Section 8). For $T(3, 4)$ with $(r, s) = (7, -5)$, a numerical computation gives a complex with zero bounding cochains, no bigon, and homology of rank 7 rather than 5 [Wue26c].

The S-complex $\tilde{C}(K; \mathbb{F}_2)$ has $d = 0$, and under (a) so does $\mathcal{P}(K)$. For an S-complex with $d = 0$, the map to the reducible is nonzero if and only if Daemi and Scaduto's invariant h is positive [DS24b, Prop. 4.15, Rem. 4.2], and h depends only on the S-chain homotopy type. Hence the conjecture implies that some x_i^- is joined to r_+ by an odd number of bigons if and only if $\delta_1 \otimes \mathbb{F}_2 \neq 0$, that is, if and only if the homomorphism $\delta_1: C_1(K; \mathbb{Z}) \rightarrow \mathbb{Z}$ takes an odd value. This condition implies $h(K) > 0$. For $T(3, 4)$ and $T(3, 5)$ it is equivalent to $h(K) > 0$. There, for each irreducible generator α of degree 1, the moduli space defining $\delta_1(\alpha)$ consists of a single instanton [DS24b, §9.3, and the same argument in §9.4], so $\delta_1(\alpha) = \pm 1$. Section 8 lists the evidence: every torus knot with nonzero pillowcase differential in [HHK18] is consistent with (a) and with the components predicted by Theorem 1.3, including $T(4, 5)$, whose bigon avoids r_+ as Hedden, Herald and Kirk point out [HHK18, §11.8], and $T(4, 7)$, with one bigon into r_+ and one between irreducible generators, exactly as Theorem 1.3 requires. For the torus knots $T(3, n)$, part (a) holds, and the pillowcase complex has the graded generators of $\tilde{C}(K)$ and a differential of rank R consisting of a single component of type δ_1 , and combined with a cobordism inequality of Daemi and Scaduto this gives part (b) for every $T(3, n)$ [Wue26c, Thm. 1.3]. Beyond torus knots, for the pretzel knots $P(-2, 3, q)$ the pillowcase complex of a decomposition obtained from that of $T(3, 5)$ by Dehn twists along the Conway sphere is again an S-complex over $\mathbb{F}_2$, but its only nonzero component is the differential d of the irreducible part, which is nonzero for $q \geq 9$ [Wue26b, Rem. 4.20].

1.4. Outline. The formula in Theorem 1.1 combines Moree's count $\|\Delta_{T(p,q)}\|_1 = 2ab - 1$, where $ap + bq = pq + 1$, with a reorganization of Litherland's lattice-point formula for the signature (Sections 2 and 3). From the formula we derive the step law and a reflection law relating $R(p, q)$ and $R(p, 2p - q)$; via Moree's count these are the recursions of Gordon, Litherland and Murasugi for the signature (Section 4). The lower bound is proved in Section 5. If p or q is even, it follows from a telescoping expression for R and a reciprocity law relating $u_p(q)$ and $u_q(p)$. If both are odd, the step and reflection laws express $R(p, q)$ through $R(p_1, p)$ with $p_1 < p$, one step of a Euclidean algorithm with even quotients, and reciprocity controls how $|u|$ changes under this step; because the reflection law enters with a minus sign, the induction carries an upper bound along with the lower bound. Theorem 1.2 follows in Section 6: the bound gives $R \geq 1$ unless $|u_p(q)| \leq 1$, and then the step law reduces $R(p, q)$ to four residues of q modulo $2p$. Section 7 proves Theorem 1.3, and Section 8 compares it with the pillowcase computations.

2. PRELIMINARIES

2.1. The S-complex of a torus knot.

Proposition 2.1. *Let K be a torus knot. The S-complex of K can be built on its traceless character variety, with $d = 0$, $\delta_2 = 0$, and $C_*(K) = C_1 \oplus C_3$, $\text{rank } C_1 = N_1$, $\text{rank } C_3 = N_3$. Over*

$\mathbb{Q}$ its differential has rank $R = \frac{1}{2}(1 + |\sigma(K)| - \|\Delta_K\|_1)$, independently of the choices made in the construction.

Proof. The traceless character variety of K consists of $|\sigma|/2$ irreducible points and one abelian point, all nondegenerate [DS24b, §9.5], and the complex $C_*(K)$ built on it has zero differential and the stated gradings [DS24a, §1, Cor. 3 and its proof]. Since $C_2 = 0$, the map $\delta_2: \mathbb{Z} \rightarrow C_{-2} = C_2$ vanishes. The homology of $\tilde{C}$ is isomorphic to $I^\natural(K)$ up to a grading shift [DS24b, Thm. 8.9], and $\text{rank } I^\natural(K) = \dim KHI(K) = \|\Delta_K\|_1$ by [KM11b, Prop. 1.4] and [LY22, Cor. 1.8]. Since $\dim H = \dim \tilde{C} - 2\text{rank } \tilde{d}$ and $\dim \tilde{C} = 1 + |\sigma|$, the formula follows; both sides depend only on K . $\square$

2.2. Signature and Alexander polynomial. Let $\Gamma = \langle p, q \rangle = \{ip + jq : i, j \in \mathbb{Z}_{\geq 0}\}$. Up to a unit, $\Delta_{T(p,q)}(t) = (1-t)(1-t^{pq})/((1-t^p)(1-t^q))$, a classical formula (see e.g. [DS24b, §9.6]), and this equals $(1-t) \sum_{\gamma \in \Gamma} t^\gamma$ [Mor14, Thm. 1], so every coefficient lies in $\{-1, 0, 1\}$. Let $a \geq 1$ be defined by $ap + bq = pq + 1$; equivalently a is the inverse of p modulo q .

Lemma 2.2 (Moree [Mor14, Lemma 1, Cor. 1]). $\|\Delta_{T(p,q)}\|_1 = 2ab - 1$.

For primes p and q this count is due to Carlitz; see [Mor14]. The signature is given by Litherland's lattice-point count [Lit79], [BO10, Prop. 1.1], where (i, j) ranges over $1 \leq i \leq p-1$ and $1 \leq j \leq q-1$:

$$\sigma(T(p, q)) = \#\{(i, j) : \frac{i}{p} + \frac{j}{q} \notin (\frac{1}{2}, \frac{3}{2})\} - \#\{(i, j) : \frac{i}{p} + \frac{j}{q} \in (\frac{1}{2}, \frac{3}{2})\},$$

Lemma 2.3. $|\sigma(T(p, q))| = (p-1)(q-1) - 4m(p, q)$.

Proof. No sum $i/p + j/q$ equals $\frac{1}{2}$ or $\frac{3}{2}$: from $2iq + 2jp \in \{pq, 3pq\}$ we get $p \mid 2i$, hence $2i = p$ and $j \in \{0, q\}$, which is excluded. The involution $(i, j) \mapsto (p-i, q-j)$ exchanges the sums below $\frac{1}{2}$ with those above $\frac{3}{2}$, so the number of sums outside $(\frac{1}{2}, \frac{3}{2})$ is twice the number m of sums below $\frac{1}{2}$. For (i, j) with $i/p + j/q < \frac{1}{2}$, both $(p-i, j)$ and $(i, q-j)$ give sums in $(\frac{1}{2}, \frac{3}{2})$, and these two injections have disjoint images; hence $\sigma \leq 0$ and $|\sigma| = (p-1)(q-1) - 4m$. Finally, the condition $2(iq + jp) < pq$ forces $2i < p$, and writing $k = p - 2i$ it reads $j < kq/(2p)$; since $p \nmid kq$, the number of such $j \geq 1$ is $\lfloor kq/2p \rfloor$, and k runs over $1 \leq k \leq p-2$ with $k \equiv p \pmod{2}$. So $m = m(p, q)$. $\square$

3. THE FLOOR-SUM FORMULA

We combine Lemmas 2.2 and 2.3 with a count of the floor sums F .

Proof of the formula in Theorem 1.1. Let $g = (p-1)(q-1)/2$, the genus of $T(p, q)$. The residues of jq modulo p for $0 \leq j \leq p-1$ are $0, \dots, p-1$, so $F(p) = \frac{1}{p}(q\binom{p}{2} - \binom{p}{2}) = g$. For $0 \leq i \leq b-1$ we have $(p-b+i)q = ap+iq-1$, so $A_{p-b+i} = a-1$ for $i=0$ and $A_{p-b+i} = a+A_i$ for $1 \leq i \leq b-1$ (as $p \nmid iq$). Summing, $F(p) - F(p-b) = F(b) + ab - 1$, that is, $F(b) + F(p-b) = g + 1 - ab$. By Lemmas 2.2 and 2.3,

$$R = \frac{1}{2}(1 + 2g - 4m - (2ab - 1)) = 1 + g - ab - 2m = F(b) + F(p-b) - 2m. \quad \square$$

Remark 3.1. In terms of the semigroup, $F(p-b)$, $F(b)$ and m are the numbers of elements of Γ in $[0, X_1]$, $[0, X_2]$ and $[0, M)$, where $X_1 = pq - p - q - bq$, $X_2 = bq - p - q - 1$ and $M = \frac{1}{2}(pq - 2p - 2q)$. Thus

$$R = \#(\Gamma \cap [0, X_1]) + \#(\Gamma \cap [0, X_2]) - 2\#(\Gamma \cap [0, M)), \quad X_1 + X_2 = 2M - 1,$$

so R compares the counting function of Γ at two points with its value at their midpoint. For $p < q$, Theorem 1.1 shows that R vanishes only when b is close to $p/2$, that is, when X_1 and X_2 are nearly symmetric about M .

4. THE STEP AND REFLECTION LAWS

Write $u = u_p(q)$; it has the parity of p . The following two recursions are, via Lemma 2.2, the recursions of Gordon, Litherland and Murasugi for the signature [GLM81, Thm. 5.2]; we derive them from the floor-sum formula.

Lemma 4.1 (Step law). *For $p \geq 2$ and $q \geq 1$ coprime, $R(p, q + 2p) = R(p, q) + \lfloor u^2/2 \rfloor$.*

Proof. The inverse b is unchanged. Each A_j increases by $2j$, so $F(n)$ increases by $n(n-1)$, and each term $\lfloor kq/2p \rfloor$ increases by k , so m increases by $\Sigma_1 = \sum k$ over $1 \leq k \leq p-2$, $k \equiv p \pmod{2}$. Using $b^2 + (p-b)^2 = \frac{1}{2}(p^2 + u^2)$, the increase of R is $\frac{1}{2}(p^2 + u^2) - p - 2\Sigma_1$. For p odd, $\Sigma_1 = \frac{1}{4}(p-1)^2$ and the increase is $\frac{1}{2}(u^2 - 1)$; for p even, $\Sigma_1 = \frac{p}{2}(\frac{p}{2} - 1)$ and the increase is $\frac{1}{2}u^2$. $\square$

Lemma 4.2 (Reflection law). *For coprime $p < q < 2p$, $R(p, q) + R(p, 2p - q) = \lfloor u^2/2 \rfloor + [p \text{ odd}]$.*

Proof. Put $q' = 2p - q$; then $b_p(q') = p - b$. For $1 \leq j \leq p-1$, $jq/p \notin \mathbb{Z}$, so $\lfloor jq/p \rfloor + \lfloor jq'/p \rfloor = 2j - 1$, and the floor sums satisfy $F_{p,q}(n) + F_{p,q'}(n) = (n-1)^2$ for $1 \leq n \leq p$. Likewise $\lfloor kq/2p \rfloor + \lfloor kq'/2p \rfloor = k - 1$, so $m(p, q) + m(p, q') = \Sigma_2 = \sum (k-1)$ over the same k . Hence the sum is $(b-1)^2 + (p-b-1)^2 - 2\Sigma_2 = \frac{1}{2}((p-2)^2 + u^2) - 2\Sigma_2$, which equals $\frac{1}{2}(u^2 + 1)$ for p odd ($\Sigma_2 = \frac{1}{4}(p-1)(p-3)$) and $\frac{1}{2}u^2$ for p even ($\Sigma_2 = (\frac{p}{2} - 1)^2$). $\square$

5. THE LOWER BOUND

5.1. p even.

Proposition 5.1. *Let $p \geq 2$ be even and $q \geq 1$ odd, and let $s = |u_p(q)|/2$. Then $R(p, q) = \sum_{i=0}^{s-1} (A_{p/2+i} - A_{p/2-s+i}) \geq s \lfloor sq/p \rfloor$. In particular $R(p, q) \geq u_p(q)^2/4$ when $q > p$.*

Proof. For p even the indices $k = 2l$ give $m(p, q) = \sum_{l=1}^{p/2-1} \lfloor lq/p \rfloor = F(p/2)$. The expression $F(b) + F(p-b)$ is symmetric under $b \leftrightarrow p-b$, and $\{b, p-b\} = \{p/2-s, p/2+s\}$. Hence $R = [F(p/2+s) - F(p/2)] - [F(p/2) - F(p/2-s)]$, which telescopes to the stated sum. Each term is at least $\lfloor sq/p \rfloor$, because $\lfloor x+y \rfloor - \lfloor x \rfloor \geq \lfloor y \rfloor$. If $q > p$ then $\lfloor sq/p \rfloor \geq s$. $\square$

5.2. Reciprocity. We extend u by $u_1(q) = -1$, which is $p - 2b$ with $p = b = 1$.

Lemma 5.2. *For coprime $n_1, n_2 \geq 1$, $n_1 u_{n_2}(n_1) + n_2 u_{n_1}(n_2) = -2$. Consequently, if $n_2 \geq 3$ and $n_1 \neq 2$, then $u_{n_2}(n_1)$ and $u_{n_1}(n_2)$ have opposite signs.*

Proof. For $n_1, n_2 \geq 2$ let $c = b_{n_2}(n_1)$ and $e = b_{n_1}(n_2)$. Then $cn_1 + en_2 \equiv 1$ modulo n_1 and modulo n_2 , and $n_1 + n_2 \leq cn_1 + en_2 \leq 2n_1n_2 - n_1 - n_2$, so $cn_1 + en_2 = n_1n_2 + 1$ (the identity $ap + bq = pq + 1$ of Section 2 for the pair (n_2, n_1)) and $n_1(n_2 - 2c) + n_2(n_1 - 2e) = -2$. If $n_1 = 1$, then $u_{n_2}(1) = n_2 - 2$ and $1 \cdot (n_2 - 2) + n_2 \cdot (-1) = -2$. For the sign: if $n_1 \neq 2$ then $u_{n_1}(n_2) \neq 0$, since $b_{n_1}(n_2) = n_1/2$ would give $n_1n_2/2 \equiv 1 \pmod{n_1}$, hence $n_1/2 \mid 1$. If $u_{n_1}(n_2) \geq 1$ then $n_1u_{n_2}(n_1) \leq -n_2 - 2 < 0$, and if $u_{n_1}(n_2) \leq -1$ then $n_1u_{n_2}(n_1) \geq n_2 - 2 > 0$. $\square$

5.3. p odd, q even.

Proposition 5.3. *Let $p \geq 3$ be odd and $q > p$ even. Then $R(p, q) \geq (u_p(q)^2 - 1)/4$.*

Proof. Put $\ell = |u_p(q)|$ and $s' = |u_q(p)|/2$. Proposition 5.1 applied to (q, p) gives $R = R(q, p) \geq s' \lfloor s'p/q \rfloor \geq 0$, which settles the case $\ell = 1$. Let $\ell \geq 3$. By Lemma 5.2, $2ps' = |qu_p(q) + 2| \geq q\ell - 2$. Since $q \geq p+1$, this gives $s' \geq \ell/2 + (\ell-2)/(2p) > \ell/2$, so $s' \geq (\ell+1)/2$; and $s'p/q \geq \ell/2 - 1/q > (\ell-1)/2$, so $\lfloor s'p/q \rfloor \geq (\ell-1)/2$. Hence $R \geq (\ell+1)(\ell-1)/4$. $\square$

5.4. **p and q odd.** This case is handled by a descent $(p, q) \mapsto (p_1, p)$, one step of a Euclidean algorithm with even quotients.

Lemma 5.4. *Let $3 \leq p < q$ be odd and coprime, and let $\rho \in (0, 2p)$ be the residue of q modulo $2p$; then $\rho \neq p$ because $p \nmid q$. Put $(p_1, \varepsilon) = (\rho, +1)$ if $\rho < p$ and $(p_1, \varepsilon) = (2p - \rho, -1)$ if $\rho > p$, and let $k \geq 1$ be such that $q = 2kp + \varepsilon p_1$. Write $e_0 = u_p(p_1)$, $e_1 = u_{p_1}(p)$, $\theta_0 = |e_0|$ and $\theta_1 = |e_1|$. Then p_1 is odd and coprime to p , $1 \leq p_1 \leq p - 2$, and:*

- (1) $|u_p(q)| = \theta_0$ and $|u_q(p)| = 2k\theta_0 + \varepsilon\theta_1$;
- (2) $\theta_1 \leq \theta_0$;
- (3) $R(p, q) = R(p_1, p) + k(\theta_0^2 - 1)/2$ if $\varepsilon = +1$, and $R(p, q) = k(\theta_0^2 - 1)/2 + 1 - R(p_1, p)$ if $\varepsilon = -1$.

For example, $(p, q) = (5, 13)$ gives $\rho = 3$, $p_1 = 3$, $\varepsilon = +1$ and $k = 1$.

Proof. Since $q \equiv \varepsilon p_1 \pmod{p}$ and $u_p(-n) = -u_p(n)$, we have $u_p(q) = \varepsilon e_0$. Lemma 5.2 for (p_1, p) gives $p_1 e_0 + p e_1 = -2$, so e_0, e_1 are odd of opposite signs, and $p_1 \theta_0 \geq p \theta_1 - 2 \geq (p_1 + 2) \theta_1 - 2 \geq p_1 \theta_1$, which is (2). Lemma 5.2 for (p, q) gives $pu_q(p) = -2 - q\varepsilon e_0 = -2k\varepsilon p e_0 + p e_1$ (substitute $q = 2kp + \varepsilon p_1$ and $p_1 e_0 = -2 - p e_1$), so $u_q(p) = -2k\varepsilon e_0 + e_1$; the two terms have the same sign if $\varepsilon = +1$ and opposite signs with $2k\theta_0 > \theta_1$ if $\varepsilon = -1$, which gives (1). For (3), $|u_p(n)| = \theta_0$ for $n \equiv \pm p_1 \pmod{p}$. If $\varepsilon = +1$ apply Lemma 4.1 k times starting from $R(p, p_1) = R(p_1, p)$. If $\varepsilon = -1$ apply it $k - 1$ times starting from $R(p, 2p - p_1)$, which equals $(\theta_0^2 + 1)/2 - R(p_1, p)$ by Lemma 4.2. $\square$

Because the reflection law enters with a minus sign, the induction must carry an upper bound alongside the lower one.

Proposition 5.5. *For odd coprime $1 \leq p < q$,*

$$\frac{u_p(q)^2 - 1}{4} \leq R(p, q) \leq \frac{u_q(p)^2 + 3}{4}.$$

Proof. Induction on p . For $p = 1$, $R(1, q) = 0$ and $u_1(q)^2 = 1$. For $p \geq 3$, take the data of Lemma 5.4; by induction $(\theta_1^2 - 1)/4 \leq R(p_1, p) \leq (\theta_0^2 + 3)/4$.

Lower bound: if $\varepsilon = +1$, $R(p, q) \geq k(\theta_0^2 - 1)/2 \geq (\theta_0^2 - 1)/4$. If $\varepsilon = -1$, $R(p, q) \geq k(\theta_0^2 - 1)/2 + 1 - (\theta_0^2 + 3)/4 = (2k - 1)(\theta_0^2 - 1)/4 \geq (\theta_0^2 - 1)/4$.

Upper bound, with $\theta = |u_q(p)|$: if $\varepsilon = +1$, then $4R(p, q) \leq \theta_0^2 + 3 + 2k(\theta_0^2 - 1)$, while $\theta^2 \geq 4k^2\theta_0^2 \geq (2k + 1)\theta_0^2 > \theta_0^2 + 2k(\theta_0^2 - 1)$. If $\varepsilon = -1$, then $4R(p, q) \leq 2k(\theta_0^2 - 1) + 4 - (\theta_1^2 - 1)$ and $\theta = 2k\theta_0 - \theta_1$, and expanding,

$$\theta^2 + 3 - [2k(\theta_0^2 - 1) + 5 - \theta_1^2] = 2(\theta_0 - \theta_1)^2 + 2(2k - 1)(k - 1)\theta_0^2 + 4(k - 1)\theta_0(\theta_0 - \theta_1) + 2(k - 1) \geq 0. \quad \square$$

Proof of the lower bound in Theorem 1.1. If p is even, this is Proposition 5.1. If p is odd then $u_p(q)$ is odd and $\lfloor u_p(q)^2/4 \rfloor = (u_p(q)^2 - 1)/4$; use Proposition 5.3 if q is even and Proposition 5.5 if q is odd. The step law is Lemma 4.1. $\square$

6. PERFECTNESS

Lemma 6.1. *For coprime $p \geq 2$ and q , $|u_p(q)| \leq 1$ if and only if $p = 2$ or $q \equiv \pm 2 \pmod{p}$.*

Proof. $u_p(q) = 0$ means $p = 2b$, which with $\gcd(b, p) = 1$ forces $p = 2$. $|u_p(q)| = 1$ means p odd and $b = (p \pm 1)/2 \equiv \pm 2^{-1} \pmod{p}$, that is, $q \equiv \pm 2 \pmod{p}$. $\square$

Lemma 6.2. *$R(2, n) = 0$ for every odd $n \geq 1$. For odd $p \geq 3$, $R(p, 2p - 2) = 1$ and $R(p, p + 2) = [p \equiv 3 \pmod{4}]$; moreover $R(p, p - 2) = [p \equiv 1 \pmod{4}]$, including $R(3, 1) = 0$.*

Proof. For $(2, n)$ the formula of Theorem 1.1 gives $R(2, n) = 0$, since $b = 1$, $F(1) = 0$ and $m(2, n) = 0$; in particular $R(p, 2) = R(2, p) = 0$. Since $b_p(2) = (p+1)/2$ and $b_p(-2) = (p-1)/2$, $|u_p(n)| = 1$ for $n \equiv \pm 2 \pmod{p}$, and Lemma 4.2 gives $R(p, 2p-2) = 1 - R(p, 2) = 1$ and $R(p, p+2) = 1 - R(p, p-2) = 1 - R(p-2, p)$. With $R(1, 3) = 0$ ($T(1, 3)$ is the unknot), induction gives $R(p, p+2) = 0$ for $p \equiv 1$ and 1 for $p \equiv 3 \pmod{4}$, and then $R(p, p-2) = 1 - R(p, p+2)$. $\square$

Proof of Theorem 1.2. If $p = 2$ then $R = 0$ by Lemma 6.2. Let $p \geq 3$. If $|u_p(q)| \geq 2$, Theorem 1.1 gives $R \geq 1$. So assume $|u_p(q)| = 1$, that is, p odd and $q \equiv \pm 2 \pmod{p}$ (Lemma 6.1). By the step law with $\lfloor u^2/2 \rfloor = 0$, $R(p, q) = R(p, q_0)$ for the residue $q_0 \in \{2, p-2, p+2, 2p-2\}$ of q modulo $2p$. Lemma 6.2 gives: $R = 0$ for $q_0 = 2$; $R = 1$ for $q_0 = 2p-2$; $R = 0$ for $q_0 = p+2$ iff $p \equiv 1 \pmod{4}$; and $R = 0$ for $q_0 = p-2$ iff $p \equiv 3 \pmod{4}$. These are the conditions stated. $\square$

Corollary 6.3. *For $n \geq 1$ coprime to 3, $R(3, n) = 0$ for $n \equiv 1, 2 \pmod{6}$ and $R(3, n) = 1$ for $n \equiv 4, 5 \pmod{6}$.*

Proof. For $p = 3$, $|u_3(n)| = 1$ for every n , so by the step law $R(3, n)$ depends only on n modulo 6, and the four residues are covered by Lemma 6.2. For $n = 4, 5$ this is [DS24b, §§9.3–9.4], and for $n \equiv 1, 2 \pmod{6}$ it is [DS24b, Prop. 9.37]. $\square$

7. COMPONENTS OF THE DIFFERENTIAL

Lemma 7.1 (after Kronheimer and Mrowka [KM14, §11]). *Let K be a torus knot, and let β_l be the rank of $I^{\natural}(K; \mathbb{Q})$ in degree $l \in \mathbb{Z}/4$. Then one of $\beta_0 - \beta_2$ and $\beta_1 - \beta_3$ is 0 and the other is ± 1 . The same holds for the ranks of the homology of $\tilde{C}(K; \mathbb{Q})$, graded as in Section 2.*

Proof. Let $\bar{\Sigma}$ be the closed surface obtained by capping off a Seifert surface of K , of genus g , in Kronheimer and Mrowka's closure of the knot complement, and let $\mu = \mu(\bar{\Sigma})$ be the corresponding operator on $KHI(K; \mathbb{Q})$. Its generalized eigenspaces $KHI(K, j)$ belong to the eigenvalues $2j$, $|j| \leq g$ [KM10a, Def. 2.5]; $KHI(K, j)$ is the summand in Alexander grading j . With respect to the canonical $\mathbb{Z}/2$ grading, the Euler characteristic of $KHI(K, j)$ is $-c_j$, where c_j is the coefficient of t^j in the symmetrized Alexander polynomial [KM10a, Thm. 1.1], [Lim10]. We give $KHI(K; \mathbb{Q})$ the relative $\mathbb{Z}/4$ grading carried over from $I^{\natural}(K; \mathbb{Q})$ along Kronheimer and Mrowka's isomorphism; they state that this isomorphism respects the mod 4 grading and that μ has degree 2 for it [KM14, §11]. Only the relative grading is used, since the conclusion is unchanged by a cyclic shift or a reversal of the grading.

- (i) For torus knots $\dim KHI(K, j) = |c_j|$ [LY22, Cor. 1.8].
- (ii) The central coefficient is $c_0 = \pm 1$. Indeed, up to a unit $\Delta_K = (1-t) \sum_{\gamma \in \Gamma} t^{\gamma}$ with $\Gamma = \langle p, q \rangle$ (Section 2), a polynomial of degree $2g$ whose coefficient of t^g is the central coefficient; so $c_0 = \pm([g \in \Gamma] - [g-1 \in \Gamma])$, and exactly one of g and $g-1$ lies in Γ , because $n \in \Gamma$ if and only if $2g-1-n \notin \Gamma$.
- (iii) For $j \neq 0$, $V_j = KHI(K, j) \oplus KHI(K, -j)$ is the generalized eigenspace of μ^2 for the eigenvalue $4j^2$. Since μ^2 has degree 0, V_j is a graded subspace, and μ restricts to an isomorphism of V_j of degree 2; so the ranks of V_j in degrees l and $l+2$ agree. Likewise $KHI(K, 0)$, the generalized kernel of μ^2 , is graded, and by (i) and (ii) it is one-dimensional.
- (iv) Summing over j , the ranks of $KHI(K; \mathbb{Q})$ in degrees l and $l+2$ agree for the pair of degrees not containing the degree of $KHI(K, 0)$, and differ by one for the other pair.

By the choice of grading the same holds for $I^{\natural}(K; \mathbb{Q})$, and by [DS24b, Thm. 8.9], which identifies the homology of $\tilde{C}(K; \mathbb{Q})$ with $I^{\natural}(K; \mathbb{Q})$ up to a shift of grading, it holds for the homology of $\tilde{C}(K; \mathbb{Q})$. $\square$

The proof uses only that $\dim KHI(K; \mathbb{Q}) = \|\Delta_K\|_1$ and that the central coefficient of Δ_K is ± 1 .

Proof of Theorem 1.3. (1) Let $\tilde{C}$ and $\tilde{C}'$ be S-complexes of K built on its character variety. They are S-chain homotopy equivalent [DS24b, Thm. 3.34]. A morphism $\tilde{C} \rightarrow \tilde{C}'$ has the block form

$$\begin{pmatrix} P & 0 & 0 \\ N & P & E_2 \\ E_1 & 0 & 1 \end{pmatrix},$$

subject to the chain-map relations of [DS24b, Def. 3.28]; the form is forced by commuting with χ [DS24b, §4.1]. The gradings force $N = 0$ (it would map $C_1 \rightarrow C'_0$ and $C_3 \rightarrow C'_2$, both zero) and $E_1 = 0$ (it would map $C_0 = 0$ to $\mathbb{Q}$), and the relations reduce to $\delta_1 = \delta'_1 P$ and $v'P = Pv + E_2 \delta_1$, where $E_2 \in C'_3$ is the image of $1 \in \mathbb{Q}$. An S-chain homotopy has degree 1, and C is concentrated in degrees 1 and 3, so its diagonal blocks vanish. Hence the diagonal blocks of a morphism and of its homotopy inverse are inverse to each other, and P is invertible. With $w = E_2$ the relations become $v' = PvP^{-1} + w \delta_1 P^{-1}$ and $\delta'_1 = \delta_1 P^{-1}$. Conversely every such pair (P, w) defines an S-isomorphism.

(2) Since $d = 0$, every element of C_* is a cycle, and $h(K) > 0$ if and only if $\delta_1 \neq 0$ by [DS24b, Prop. 4.15].

(3) The map $w \delta_1$ sends C_1 to C_3 and vanishes on $\ker \delta_1$, so it changes neither v_{31} nor $v_{13}|_{\ker \delta_1}$, and P conjugates both; hence their ranks are invariant. With $d = 0$ and $\delta_2 = 0$, $\tilde{d}$ is determined by the map $(v, \delta_1): C \rightarrow C \oplus \mathbb{Q}$, whose rank is $[\delta_1 \neq 0] + \text{rank}(v|_{\ker \delta_1})$, and $\text{rank}(v|_{\ker \delta_1}) = \text{rank } v_{31} + \text{rank}(v_{13}|_{\ker \delta_1})$ because $C_3 \subset \ker \delta_1$ and the two pieces land in the complementary summands C_1 and C_3 . Let $R_3 = \text{rank } v_{31}$, and let R_1 be the rank of $(\delta_1, v_{13}): C_1 \rightarrow \mathbb{Q} \oplus C_3$; these are the ranks of the components of $\tilde{d}$ of degrees $3 \rightarrow 2$ and $1 \rightarrow 0$. Then $R_1 + R_3 = R$ and $R_1 = [h > 0] + \text{rank}(v_{13}|_{\ker \delta_1})$ by (2). The complex $\tilde{C}$ has rank vector $(N_3 + 1, N_1, N_1, N_3)$ in degrees 0, 1, 2, 3, so its homology has rank vector

$$\beta = (N_3 + 1 - R_1, N_1 - R_1, N_1 - R_3, N_3 - R_3).$$

Put $D = R_3 - R_1$. Then $\beta_0 - \beta_2 = 1 - (N_1 - N_3) + D$ and $\beta_1 - \beta_3 = (N_1 - N_3) + D$. Since $|\sigma|$ is even, $N_1 - N_3 \in \{0, 1\}$, so these two differences are D and $D + 1$ in some order. By Lemma 7.1 one of them is 0 and the other is ± 1 , so $D \in \{0, -1\}$. With $R_1 + R_3 = R$ this gives $R_3 = \lfloor R/2 \rfloor$ and $R_1 = \lceil R/2 \rceil$.

(4) If $h = 0$ then $\delta_1 = 0$ by (2), the transformation in (1) is conjugation by P , and v has rank R . With $d = \delta_1 = \delta_2 = 0$, Daemi and Scaduto's complex computing $\hat{I}(K; \mathbb{Q})$ is $C_{*-1} \oplus \mathbb{Q}[x]$, with x acting by v on the first summand [DS24b, §4.3, Lemma 4.11], so (C, v) is the $\mathbb{Q}[x]$ -torsion of $\hat{I}(K; \mathbb{Q})$, up to a shift of degree, and it is an invariant of K [DS24b, Thm. 1.4]. $\square$

Proof of Corollary 1.4. Substitute $R_1 = \lceil R/2 \rceil$ and $R_3 = \lfloor R/2 \rfloor$ into β . $\square$

Example 7.2. For $T(4, 5)$, $\sigma = -8$ and $N_1 = N_3 = 2$, so $\tilde{C}$ has rank vector $(3, 2, 2, 2)$ in degrees 0, 1, 2, 3, and since $\sigma \equiv 0 \pmod{4}$ no shift is needed to compare with $I^\natural$. Kronheimer and Mrowka showed that $I^\natural(T(4, 5))$ has rank vector $(2, 1, 2, 2)$ [KM14, §11]; see also [HHK14, §12.5]. A differential of degree -1 with these chain and homology ranks has a single nonzero component, of rank one, from degree 1 to degree 0. The maps of degree $1 \rightarrow 0$ are δ_1 and v_{13} , and $\delta_1 = 0$ because $h(T(4, 5)) = 0$. Hence v_{13} has rank one and $v_{31} = 0$. This is the instanton counterpart of the single pillowcase bigon of $T(4, 5)$, which also lowers the grading from 1 to 0 and avoids the reducible generator [HHK18, §11.8].

Remark 7.3. The rank of v itself is not an invariant when $\delta_1 \neq 0$: for $T(3, 4)$ and $T(3, 5)$, the representatives with $v = 0$ and with $v = w\delta_1$ ($w \in C_3$, $w \neq 0$) are S-isomorphic, by Theorem 1.3(1) with $P = 1$. Daemi and Scaduto observe this freedom in the choice of splitting [DS24b, Rem. 4.4].

Remark 7.4. For $n \equiv 4, 5 \pmod{6}$, $h(T(3, n)) = 1$. Write $\Upsilon_n = \Upsilon_{T(3, n)}(1)$ and $\sigma_n = \sigma(T(3, n))$. By [FK17, Prop. 2.2], $\Upsilon_{n+3} = \Upsilon_n + \Upsilon_4$ for $n \geq 1$, and $\Upsilon_4 = -2$, since $h(T(3, 4)) = 1$ [DS24a, §4.1, Prop. 10] and $\sigma_4 = -6$. Since also $\sigma_{n+6} = \sigma_n - 8$ [GLM81, Thm. 5.2], $h(T(3, n)) = \Upsilon_n - \sigma_n/2$ depends only on n modulo 6. Finally $\Upsilon_5 = \Upsilon_2 + \Upsilon_4 = -3$ and $\sigma_5 = -8$, so $h(T(3, 5)) = 1$. By Theorem 1.3(3), $v_{31} = 0$ and v_{13} vanishes on $\ker \delta_1$, so $v_{13} = w\delta_1$ for some $w \in C_3$, and the S-isomorphism $(1, -w)$ makes $v = 0$: up to S-isomorphism over $\mathbb{Q}$ the differential of $T(3, n)$ is δ_1 alone.

8. COMPARISON WITH THE PILLOWCASE

Hedden, Herald and Kirk decompose $T(p, q)$ along a Conway sphere bounding a ball in which the knot is a trivial tangle. The trivial tangle carries the earring and gives a figure-eight curve L_0 in the pillowcase; the other tangle gives, after a small holonomy perturbation, an immersed curve L_1 [HHK18, §§10–11]. Table 1 compares their computations with Theorem 1.3. After the grading is lowered by σ , a bigon into r_+ corresponds to δ_1 (degree $1 \rightarrow 0$); any other bigon from grading $\sigma + 3$ to $\sigma + 2$ corresponds to v_{31} (degree $3 \rightarrow 2$); and any other bigon from grading $\sigma + 1$ to σ corresponds to $v_{13}|_{\ker \delta_1}$ (degree $1 \rightarrow 0$).

knot	R	h	$\sigma \bmod 4$	bigons in [HHK18], gradings of [HHK18]	component of $\tilde{d}$
$T(3, 4)$	1	1	2	$x_1^- \rightarrow r_+$, $3 \rightarrow 2$ (§11.6)	δ_1
$T(3, 5)$	1	1	0	$x_1^- \rightarrow r_+$, $1 \rightarrow 0$ (§11.7)	δ_1
$T(4, 5)$	1	0	0	one bigon avoiding r_+ , $1 \rightarrow 0$ (§11.8)	v_{13}
$T(4, 7)$	2	1	2	$x_3^- \rightarrow r_+$, $3 \rightarrow 2$; $x_1^- \rightarrow x_2^+$, $1 \rightarrow 0$ (§11.9)	δ_1 ; v_{31}
$T(5, 11)$	4	0	0	four bigons on closed components (§11.1)	v_{31} , v_{13} , rank 2 each

TABLE 1. Torus knots with nonzero pillowcase differential in [HHK18]. The gradings in the fifth column are those of [HHK18]; subtracting σ gives the degrees of the components in the last column, which are determined by Theorem 1.3 and Example 7.2. In each case a bigon ends at r_+ exactly when $h > 0$, and the rank of the pillowcase differential equals R . The knots $T(3, 7)$, $T(5, 7)$, $T(5, 12)$ and $T(5, 17)$ of [HHK18, §11] have $R = 0$ and no differential.

For $T(5, 11)$, the curve L_1 consists of an arc, which contributes r_+ , and four closed components [HHK18, §11.1]. The four bigons lie on two of the closed components, two on each, and cancel $8 = 25 - 17$ of the 25 generators in pairs; as a linear map the differential has rank $4 = R$. On each of these components the two bigons run from some x_i^- to some x_j^+ with $j \neq i$ [HHK18, §3.9], as Conjecture 1.5(a) requires; the absolute gradings of closed components are fixed by convention in [HHK18], so this row does not test the split between v_{31} and v_{13} . For $T(4, 7)$, Theorem 1.3 gives rank $v_{31} = 1$ and $v_{13}|_{\ker \delta_1} = 0$, which is what the two bigons show. (Hedden, Herald and Kirk name the second pair x_2^+, x_1^- ; its direction $x_1^- \rightarrow x_2^+$ follows from the gradings.) Over $\mathbb{F}_2$, the bigon into r_+ for $T(4, 7)$ also predicts that δ_1 is nonzero modulo 2, which is not known. The pillowcase complex depends on the perturbation: for larger perturbation the pair joined by the bigon into r_+ cancels and the complex of $T(3, 4)$ becomes perfect [HHK18, §11.10, Prop. 11.1]. This is consistent with $h(T(3, 4)) = 1$: once that pair has cancelled, the complex no longer carries the S-structure of Conjecture 1.5, whereas h is an invariant of the S-complex. The conjecture therefore concerns the regime of small perturbations. With the earring on the

other tangle the pillowcase homology of $T(4, 5)$ has the wrong rank [CHKK22, §10.3], so the choice of decomposition matters as well, and so does the choice of (r, s) within the family of [HHK18, §10] [Wue26c]. A proof would amount to a chain-level form of the knot Atiyah–Floer correspondence of [HHK18, CHKK22] for torus knots; Cazassus, Herald, Kirk and Kotelskiy already suggest viewing the earring as a counterpart of the equivariant variable x [CHKK22, §11.6].

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BERND JOHANNES WUEBBEN, NEW YORK, NY, wuebben@gmail.com